

Resume

Name: Xiaoyu Luo Date of Birth: August 2004 Height: 175 cm
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Education

South China Normal University - Software Engineering - Bachelor's Degree 2022-09
~ 2026-07

The Hong Kong University of Science and Technology - Artificial Intelligence and
Entrepreneurship (AIE) - Full-time Postgraduate Student 2026-09 ~ Present

Academic Performance (South China Normal University): GPA 3.33

Major Courses: Fundamentals of Python Programming, Computer Architecture, Object-Oriented Programming, Introduction to Software Engineering, Java Programming, Data Structures and Algorithms, Algorithm Design and Analysis, Operating Systems, Database Systems: Principles and Practice, Web Application Development, Linux Systems, Computer Networks, Software System Design, Software Testing, Fundamentals of Artificial Intelligence, Network Security, Data Mining, Robotics, Software Design, Distributed Systems, Knowledge Representation, Natural Language Processing, Computational Intelligence.

Project Experience

Server Operations (Internship) 2024-10 ~ 2024-12

Completed a one-month internship at Guangzhou Wanglv Internet Technology Co., Ltd., mainly responsible for server operations, including optimizing back-end server algorithms and writing data migration scripts for a bastion host. During the internship, became familiar with building and operating back-end servers using Python and FastAPI, the principles and application of Nginx proxying, and basic Linux operations and scripting. Also gained an understanding of the software project life cycle, from requirements analysis through code submission and review to deployment, as well as workplace communication and collaboration.

Online Collaborative Note-Taking Platform - Back-End Server and Database Setup and Operations 2025-02 ~ 2025-5

Previous back-end development was based mainly on the lightweight FastAPI framework, which did not effectively support interaction between the business layer and the database. This project therefore used Spring Boot, a Java-based framework for medium-to-large applications. Starting with Java Web fundamentals and progressing to a complex Spring Boot framework provided a broad understanding of the basic enterprise development process. The project adopted a layered architecture, dividing the server into presentation,

business, and data layers for receiving front-end data, processing business logic, and creating, reading, updating, and deleting user data.

Binary Fuzzing Agent (Internship) - Full Development 2025-07 ~ 2025-9

Completed a three-month internship at Beijing Yanling Wangwei Intelligent Technology Co., Ltd., primarily responsible for developing a binary fuzzing agent. During this period, learned taint analysis and fuzzing techniques in vulnerability detection, mainly targeting binary executable programs. Later work focused on developing an AFL-based automated fuzzing agent and applying LLMs to initial test-case generation, seed mutation, result analysis, and report generation.

Airport Runway Scheduling Optimization Algorithm Design - Capstone Project

2026-02 ~ 2026-06

Designed an airport runway scheduling solution based on the ant colony algorithm. Under constraints including runway count, minimum safety separation, and scheduled takeoff and landing times, the solution generates an optimized schedule with the objective of minimizing total flight delay. It uses a front-end/back-end architecture: the front end collects parameters such as runway count, safety separation, and scheduled flight times and validates their values and compliance. The back end runs the optimization algorithm and returns the results to the front end, which displays the optimized takeoff and landing sequence and scheduling recommendations.

Skills and Strengths

Computer Skills: Proficient in the basic syntax of Python, Java, C, and other programming languages; able to read basic assembly language and write simple assembly code; experienced with database operations and proficient in MySQL syntax; systematically trained in basic Linux operations and able to write basic scripts; experienced in Web application development, proficient in native HTML, CSS, and JavaScript, and familiar with Web development using the Vue framework; proficient in back-end server architectures based on FastAPI and Spring Boot.

Teamwork: Strong communication skills and able to maintain effective working relationships with other team members. Project experience has provided an initial understanding of task allocation, version control, and collaborative division of work.

Language Skills: IELTS score of 7.0; able to read English texts without difficulty and communicate confidently in English.

Self-Assessment

I approach tasks conscientiously and responsibly and complete them ahead of schedule without procrastination. I have strong self-learning ability and experience and can use a wide range of online resources to solve problems. I have developed my own AI-assistant toolchain and, in addition to professional skills, have learned basic video, image, audio, and text editing. I have acquired many foundational software engineering skills

and place strong emphasis on practical application. I hope to gain more hands-on development experience and learn how the industry operates.

My undergraduate studies covered a wide range of topics but often only at an introductory level. I hope to use internships to study the high-level logic and underlying principles of project development more deeply, including data processing and software development.